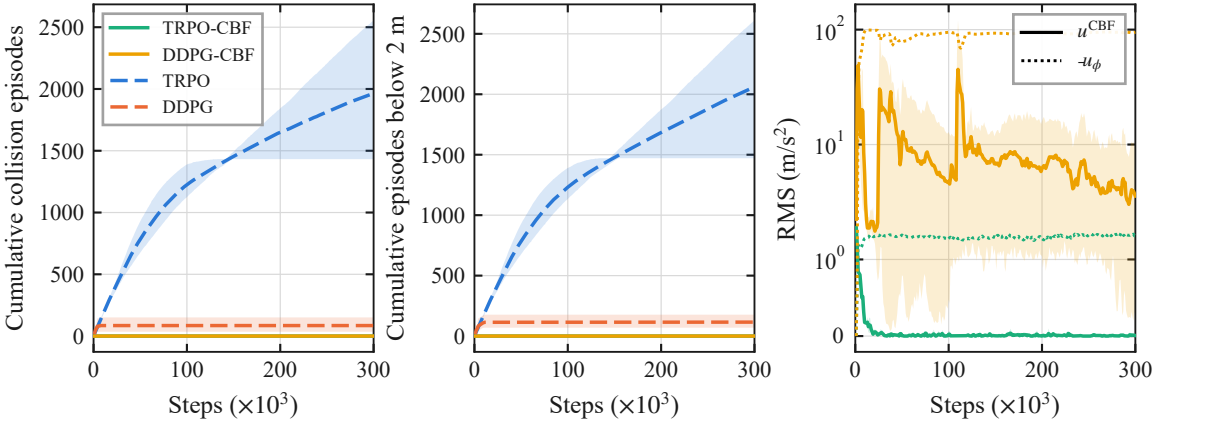

(a) Collisions
Cumulative collision episodes
TRPO-CBF
DDPG-CBF
TRPO
DDPG
0
500
1000
1500
2000
2500
0
100
200
300
Steps (×10^3)
(b) Safety-distance exits
Cumulative episodes below 2 m
0
500
1000
1500
2000
2500
0
100
200
300
Steps (×10^3)
(c) CBF correction / compensator
RMS (m/s^2)
u^CBF
-u_φ
10^2
10^1
10^0
0
100
200
300
Steps (×10^3)